Aligning Behavioral and Internal Semantic Geometry in LLMs

Louis Schiekiera[1,2], Max Zimmer[3], Christophe Roux[3], Sebastian Pokutta[3,4], Fritz Günther[1]

[1] HU Berlin, [2] FU Berlin, [3] Zuse Institute Berlin, [4] TU Berlin

Behavioral Clones Workshop

Max Planck Institute for Human Development, Berlin

2026-05-20

Behavior as a Measurement Instrument

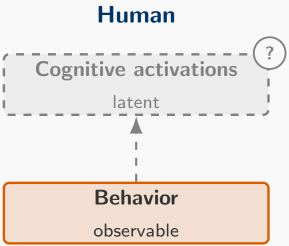
A classic cognitive-science question (Baker et al., 2009)

If you only observe a system's external **behavior**, how much can you recover about the structure of its internal **cognitive activations**?

Behavior as a Measurement Instrument

A classic cognitive-science question (Baker et al., 2009)

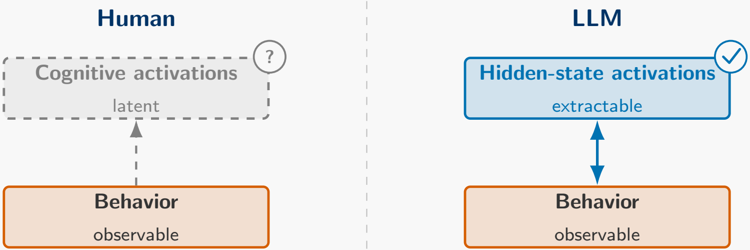
If you only observe a system's external **behavior**, how much can you recover about the structure of its internal **cognitive activations**?



Behavior as a Measurement Instrument

A classic cognitive-science question (Baker et al., 2009)

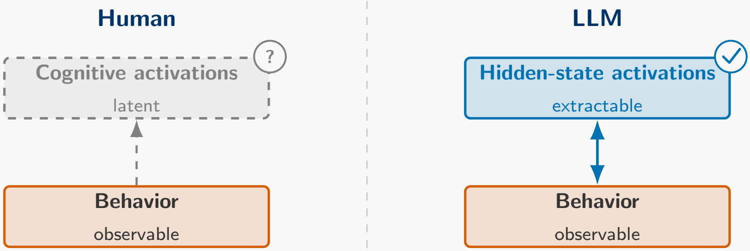
If you only observe a system's external **behavior**, how much can you recover about the structure of its internal **cognitive activations**?



Behavior as a Measurement Instrument

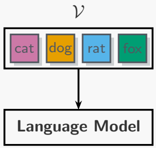
A classic cognitive-science question (Baker et al., 2009)

If you only observe a system's external **behavior**, how much can you recover about the structure of its internal **cognitive activations**?



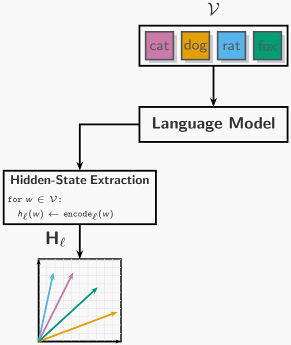
- **This work**: use psycholinguistic tasks to create a behavioral similarity structure that recovers an LLM's hidden-state similarity structure.

Conceptual Approach: Two Paths to the Same Geometry



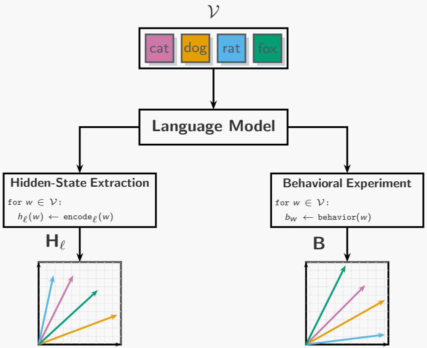
Conceptual Approach: Two Paths to the Same Geometry

- **Path A (internal)**: extract hidden states $\to$ similarity $\mathbf{S}^{\text{hid}}$.



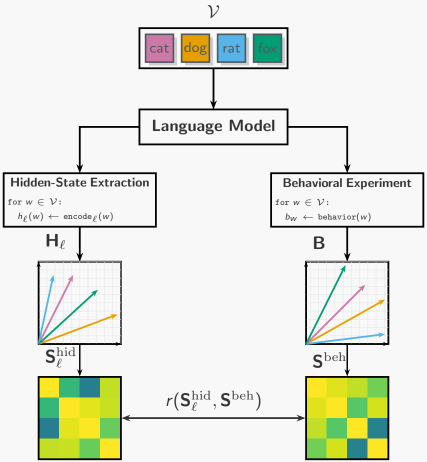
Conceptual Approach: Two Paths to the Same Geometry

- **Path A (internal)**: extract hidden states $\rightarrow$ similarity $\mathbf{S}^{\mathrm{hid}}$.
- **Path B (external)**: run behavioral tasks $\rightarrow$ similarity $\mathbf{S}^{\mathrm{beh}}$.



Conceptual Approach: Two Paths to the Same Geometry

- **Path A (internal)**: extract hidden states $\rightarrow$ similarity $\mathbf{S}^{\text{hid}}$.
- **Path B (external)**: run behavioral tasks $\rightarrow$ similarity $\mathbf{S}^{\text{beh}}$.
- **Compare**: representational similarity between $\mathbf{S}^{\text{hid}}$ and $\mathbf{S}^{\text{beh}}$.

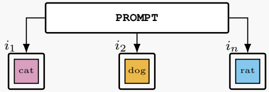


Behavioral Paradigm: Free Association (FA)

PROMPT

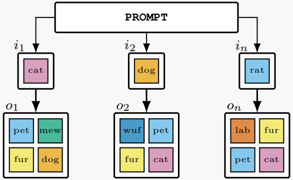
Behavioral Paradigm: Free Association (FA)

- One **input** word i.



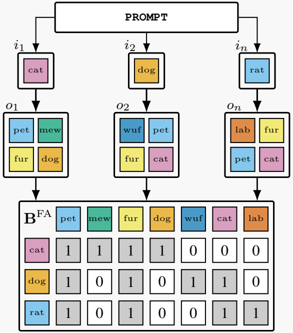
Behavioral Paradigm: Free Association (FA)

- One **input** word i.
- Pick five **output** words o that come to mind.



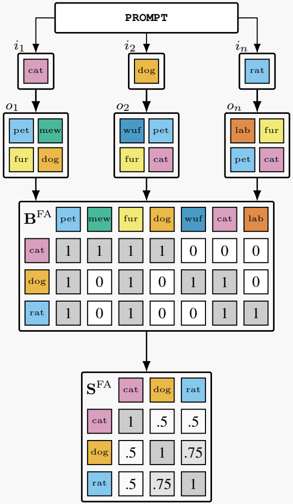
Behavioral Paradigm: Free Association (FA)

- One **input** word i.
- Pick five **output** words o that come to mind.
- Create input-output matrix $\mathbf{B}^{\mathrm{FA}}$ from i and o.



Behavioral Paradigm: Free Association (FA)

- One **input** word i.
- Pick five **output** words o that come to mind.
- Create input-output matrix $\mathbf{B}^{\mathrm{FA}}$ from i and o.
- Compute behavioral similarity matrix $\mathbf{S}^{\mathrm{FA}}$: cosine between cue rows.

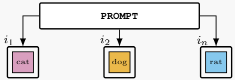


Behavioral Paradigm: Forced Choice (FC)

PROMPT

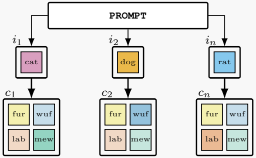
Behavioral Paradigm: Forced Choice (FC)

- One **input** word i.



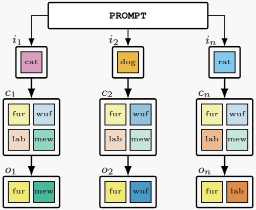
Behavioral Paradigm: Forced Choice (FC)

- One **input** word i.
- Sixteen **candidate** words c.



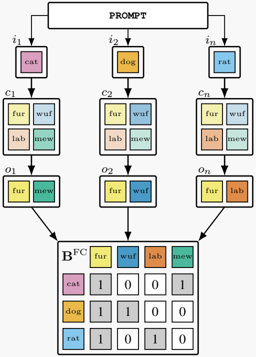
Behavioral Paradigm: Forced Choice (FC)

- One **input** word i.
- Sixteen **candidate** words c.
- Pick two **output** words o.



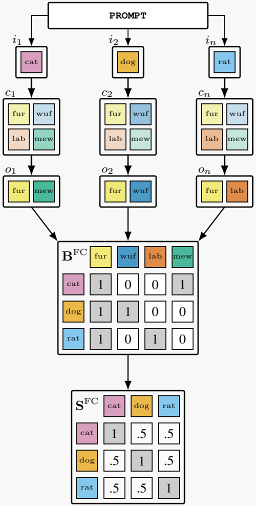
Behavioral Paradigm: Forced Choice (FC)

- One **input** word i.
- Sixteen **candidate** words c.
- Pick two **output** words o.
- Create input-output matrix $\mathbf{B}^{\mathrm{FC}}$ from i and o.

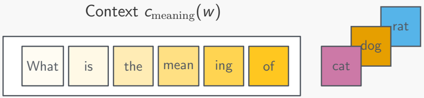


Behavioral Paradigm: Forced Choice (FC)

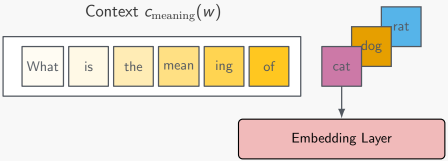
- One **input** word i.
- Sixteen **candidate** words c.
- Pick two **output** words o.
- Create input-output matrix $\mathbf{B}^{\mathrm{FC}}$ from i and o.
- Compute behavioral similarity matrix $\mathbf{S}^{\mathrm{FC}}$: cosine between cue rows.



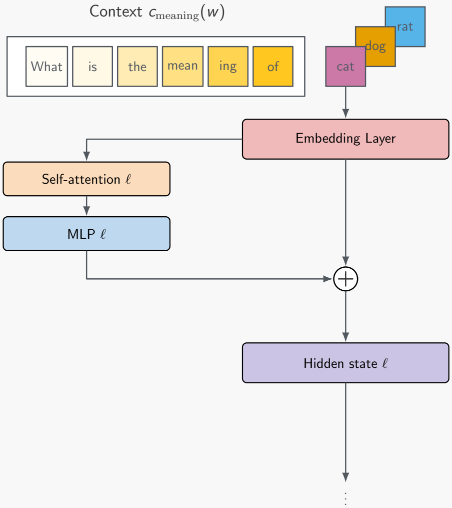
Hidden-State Extraction: The Meaning Prompt



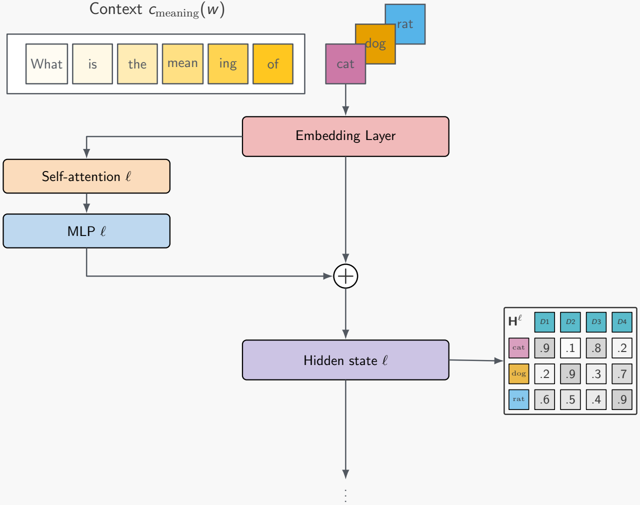
Hidden-State Extraction: The Meaning Prompt



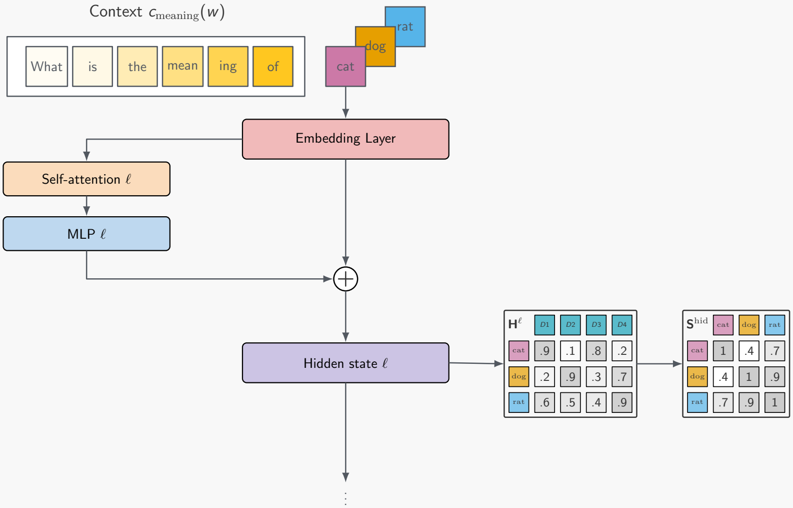
Hidden-State Extraction: The Meaning Prompt



Hidden-State Extraction: The Meaning Prompt



Hidden-State Extraction: The Meaning Prompt



Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)
- **Behavioral Trials:** 17.5M+

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)
- **Behavioral Trials:** 17.5M+
- **Baselines:** FastText, BERT, cross-model consensus

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)
- **Behavioral Trials:** 17.5M+
- **Baselines:** FastText, BERT, cross-model consensus

How we compare behavioral and hidden-state geometry

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)
- **Behavioral Trials:** 17.5M+
- **Baselines:** FastText, BERT, cross-model consensus

How we compare behavioral and hidden-state geometry

1. **RSA**: pairwise correlation of the hidden-state and behavioral similarity matrices (Kriegeskorte et al., 2008).

Scale of the Study

Model	n_{params}	n_{layers}	d_{model}
`Falcon3-10B-Instruct`	10B	40	3072
`gemma-2-9b-it`	9B	42	3584
`Llama-3.1-8B-Instruct`	8B	32	4096
`Mistral-7B-Instruct-v0.2`	7B	32	4096
`Mistral-Nemo-Instruct-2407`	12B	40	5120
`phi-4`	14B	40	5120
`Qwen2.5-7B-Instruct`	7B	28	3584
`rnj-1-instruct`	8B	32	4096

n_{params} = parameters (B); n_{layers} = layers; d_{model} = hidden-state width.

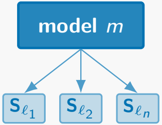
- **Models:** 8 instruction-tuned LLMs
- **Vocab:** 5,000 frequent SUBTLEX-US nouns (Brysbaert et al., 2012)
- **Behavioral Trials:** 17.5M+
- **Baselines:** FastText, BERT, cross-model consensus

How we compare behavioral and hidden-state geometry

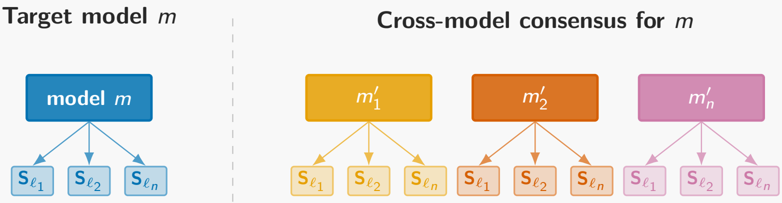
1. **RSA**: pairwise correlation of the hidden-state and behavioral similarity matrices (Kriegeskorte et al., 2008).
2. **Regression**: predict hidden-state similarity from behavior beyond baselines (Hoerl and Kennard, 1970).

Cross-Model Consensus Baseline

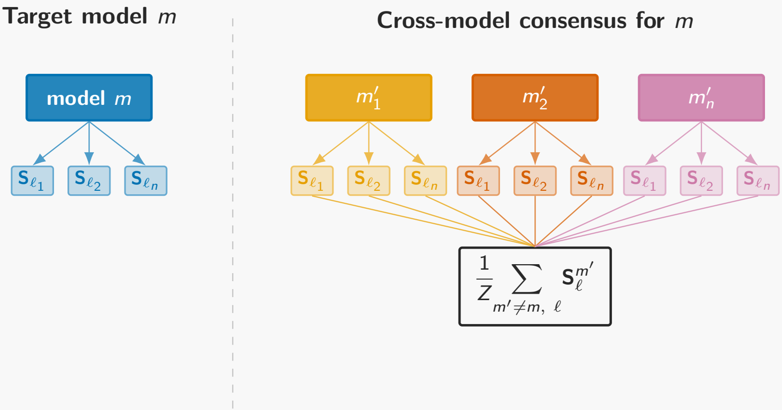
Target model m



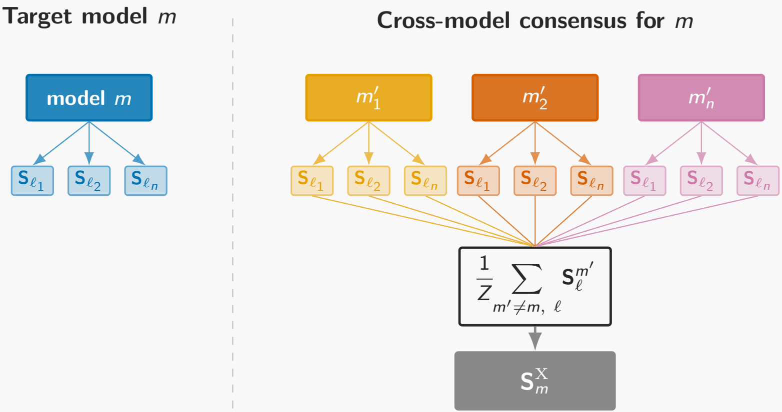
Cross-Model Consensus Baseline



Cross-Model Consensus Baseline



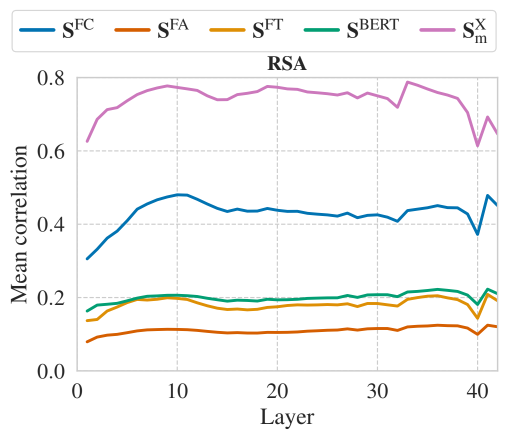
Cross-Model Consensus Baseline



Cross-model consensus: shared representational similarity structure across models (Huh et al., 2024).

Result 1: FC Aligns; FA Does Not

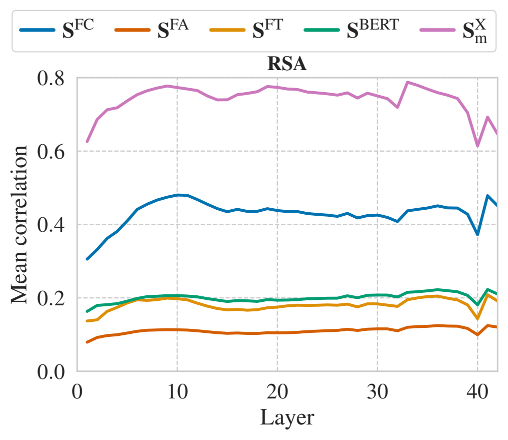
- **Universal structure dominates**: cross-model consensus aligns most strongly (Huh et al., 2024; Kaushik et al., 2025).



RSA vs. five reference geometries. Mean across 8 models and 4 extraction contexts.

Result 1: FC Aligns; FA Does Not

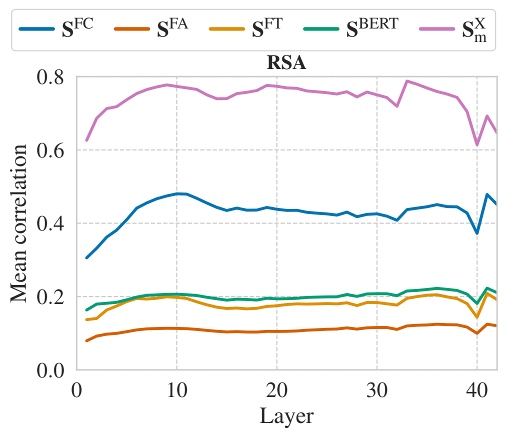
- **Universal structure dominates**: cross-model consensus aligns most strongly (Huh et al., 2024; Kaushik et al., 2025).
- **FC** $\gg$ **FA**: forced choice aligns substantially more with hidden-state geometry than free association.



RSA vs. five reference geometries. Mean across 8 models and 4 extraction contexts.

Result 1: FC Aligns; FA Does Not

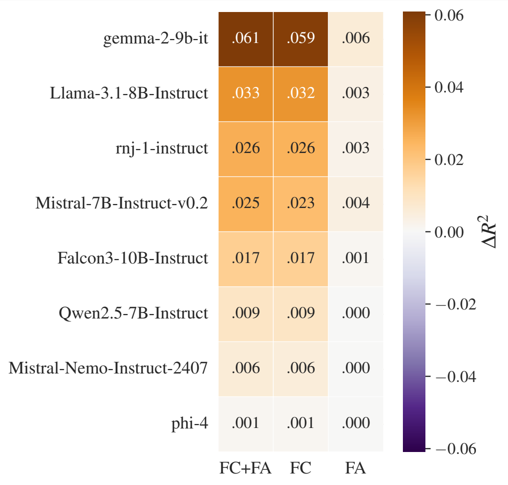
- **Universal structure dominates**: cross-model consensus aligns most strongly (Huh et al., 2024; Kaushik et al., 2025).
- **FC $\gg$ FA**: forced choice aligns substantially more with hidden-state geometry than free association.
- **Behavior beats lexical baselines**: $\mathbf{S}^{\mathrm{FC}}$ alignment exceeds FastText and BERT.



RSA vs. five reference geometries. Mean across 8 models and 4 extraction contexts.

Result 2: Behavior Predicts Hidden States on Unseen Words

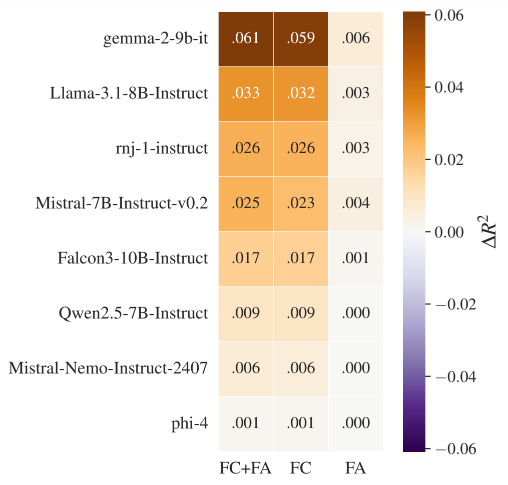
- **FC adds signal** on top of (FastText + BERT + cross-model) for nearly all models and layers (mean $\Delta R^2 = .02$).



Incremental test R^2 for behavioral predictors on top of baseline.

Result 2: Behavior Predicts Hidden States on Unseen Words

- **FC adds signal** on top of (FastText + BERT + cross-model) for nearly all models and layers (mean $\Delta R^2 = .02$).
- **FA adds no incremental signal** beyond shared cross-model structure (mean $\Delta R^2 = .00$).



Incremental test R^2 for behavioral predictors on top of baseline.

Takeaway

Bottom line

Behavior can recover a meaningful portion of internal geometry, but the measurement instrument matters.

Takeaway

Bottom line

Behavior can recover a meaningful portion of internal geometry, but the measurement instrument matters.

1. **Constrained > open-ended.** FC's dense, well-conditioned matrix $\mathbf{B}^{\mathrm{FC}} \in \mathbb{R}^{5000\times5000}$ recovers geometry; FA's sparse, heavy-tailed $\mathbf{B}^{\mathrm{FA}} \in \mathbb{R}^{5000\times\sim15000}$ does not.

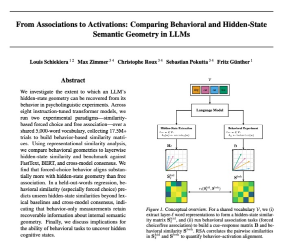
Takeaway

Bottom line

Behavior can recover a meaningful portion of internal geometry, but the measurement instrument matters.

1. **Constrained $>$ open-ended.** FC's dense, well-conditioned matrix $\mathbf{B}^{\mathrm{FC}} \in \mathbb{R}^{5000 \times 5000}$ recovers geometry; FA's sparse, heavy-tailed $\mathbf{B}^{\mathrm{FA}} \in \mathbb{R}^{5000 \times \sim 15000}$ does not.
2. **Discrete behavior alone captures internal geometry without model access.** No logits or weights required. This opens a potential path toward interpretability of closed models.

Preprint, Data, Code



From Associations to Activations: Comparing Behavioral and Hidden-State Semantic Geometry in LLMs

Louis Schiekiera [1,2] Max Zimmer [3,4] Christophe Roux [3,4] Sebastian Pokutta [3,4] Fritz Günther [1]

Abstract

We investigate the extent to which an LLM's hidden-state geometry can be recovered from its behavior in psycholinguistic experiments. Across eight instruction-tuned transformer models, we run two experimental paradigms—similarity-based forced choice and free association—over a shared 5,000-word vocabulary, collecting 17.5M+ trials to build behavior-based similarity matrices. Using representational similarity analysis, we compare behavioral geometries to layerwise hidden-state similarity and benchmark against FastText, BERT, and cross-model consensus. We find that forced-choice behavior aligns substantially more with hidden-state geometry than free association. In a held-out-words regression, behavioral similarity (especially forced choice) predicts unseen hidden-state similarities beyond lexical baselines and cross-model consensus, indicating that behavior-only measurements retain recoverable information about internal semantic geometry. Finally, we discuss implications for the ability of behavioral tasks to uncover hidden cognitive states.

Figure 1. Conceptual overview. For a shared vocabulary $\mathcal{V}$, we (i) extract layer-ℓ word representations to form a hidden-state similarity matrix $\mathbf{S}_\ell^{\text{hid}}$, and (ii) run behavioral association tasks (forced choice/free association) to build a cue–response matrix $\mathbf{B}$ and behavioral similarity $\mathbf{S}^{\text{beh}}$. RSA correlates the pairwise similarities in $\mathbf{S}_\ell^{\text{hid}}$ and $\mathbf{S}^{\text{beh}}$ to quantify behavior–activation alignment.

Scan for the preprint

Preprint (accepted to ICML 2026)
arxiv.org/pdf/2602.00628

Dataset
huggingface.co/datasets/schiekiera/llm-association-geometry

Code
github.com/schiekiera/llm-association-geometry

Thank you for your attention!

References

Baker, C. L., Saxe, R., and Tenenbaum, J. B. (2009). Action understanding as inverse planning. *Cognition*, 113(3):329–349.

Brysbaert, M., New, B., and Keuleers, E. (2012). Adding part-of-speech information to the SUBTLEX-US word frequencies. *Behavior research methods*, 44(4):991–997.

Hoerl, A. E. and Kennard, R. W. (1970). Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67.

Huh, M., Cheung, B., Wang, T., and Isola, P. (2024). The platonic representation hypothesis. *arXiv preprint arXiv:2405.07987*.

Kaushik, P., Chaudhari, S., Vaidya, A., Chellappa, R., and Yuille, A. (2025). The Universal Weight Subspace Hypothesis.

References ii

Kriegeskorte, N., Mur, M., and Bandettini, P. A. (2008). Representational similarity analysis-connecting the branches of systems neuroscience. *Frontiers in systems neuroscience*, 2:249.